

CSCE 650 VIRTUAL REALITY · MODULE 12 · LECTURE NOTES

Augmented Reality and Passthrough

Registering the virtual to the real.

MODULE OUTCOMES

- Compare optical and video see-through.
- Explain registration and why it is AR's hard problem.
- Explain the occlusion problem and the approaches to it.
- Explain lighting estimation and why it matters.
- Identify the AR-specific interaction and social issues.

1 Two kinds of see-through

	Optical see-through	Video see-through (passthrough)
Method	A transparent combiner — a half-silvered mirror or waveguide — through which the real world is seen directly, with virtual light added.	Cameras capture the real world; the display shows a composite of camera feed and virtual content.
Real world latency	Zero. It is light from the actual world.	Nonzero — capture, processing, and display, typically tens of milliseconds.
Real world quality	Full acuity and dynamic range.	Limited by the cameras — resolution, noise, dynamic range.
Virtual content	Additive only. See below.	Anything, including pure black and full occlusion.
Field of view	Typically narrow — the classic limitation of AR glasses, and difficult to widen.	As wide as the headset's display.
Current status	Research and specialist devices.	What current consumer headsets do , and it works considerably better than most people expect.

OPTICAL SEE-THROUGH CANNOT RENDER BLACK

A combiner adds light to what is already reaching the eye. It has no mechanism for subtracting light.

So a virtual object can only ever be brighter than the real background behind it. The effect is exactly like a reflection in a window: you see the reflected image superimposed on the view through the glass, and the reflection is always translucent.

Dark virtual objects are therefore impossible, and all virtual content is somewhat transparent, with real detail showing through.

Which means a virtual object can never properly occlude a real one. Module 02 established that occlusion is the strongest depth cue the visual system has — so this is a fundamental limitation on the realism achievable, not an image-quality issue that better components will resolve. **It is the main reason video passthrough has won for the present generation,** despite its own disadvantages.

2 Registration

REGISTRATION ERROR IS OBVIOUS IN AR AND INVISIBLE IN VR

In VR there is no external reference. If the entire virtual world is positioned two centimetres from where it should be, nothing reveals it — the user has nothing to compare against, and the error is simply where the world is.

In AR the real world is right there, continuously, as a reference. A virtual object that should be resting on a real table either is resting on it or is visibly floating above or sinking into it.

The tolerances are correspondingly tight: sub-centimetre positional accuracy and sub-degree orientation accuracy are needed for content at arm's length to read as genuinely attached to the world.

And latency converts directly into registration error. Any lag between head motion and the virtual layer's update makes virtual content swim relative to the real world — and because the real world is the reference, the swimming is immediately visible. **This is the characteristic AR artefact,** and it is why AR's latency requirements are even tighter than VR's.

- **Accurate tracking** — the problem of Module 05, with a substantially tighter tolerance and no opportunity to hide error.
- **Scene understanding.** Where are the floors, walls, tables, and objects? Virtual content that must relate to real geometry requires that geometry to be known, which means plane detection, mesh reconstruction, and object recognition.
- **Low latency for the virtual layer specifically,** which must stay locked to a real world that — in video passthrough — has a latency of its own. Matching the two matters as much as minimising either.
- **Camera and display calibration,** so that what the cameras capture maps correctly onto what the eye sees through the optics. Errors here produce a systematic offset that no amount of tracking accuracy corrects.

- **Persistence across sessions.** Content placed on a real table yesterday should still be on that table today, which requires recognising the space again — relocalisation against a saved map. **Persistent anchors are what make AR useful rather than a demonstration**, and they are considerably harder than they appear.

OCCLUSION: THE VIRTUAL MUST GO BEHIND THE REAL

A virtual ball rolled under a real table must disappear under it. If it continues to draw on top of the table, the illusion that it occupies the room fails instantly and completely.

This requires knowing the depth of the real world, per pixel, in real time — a considerably harder requirement than knowing where the planes are.

The approaches each fail where it matters. A dedicated depth sensor is accurate and has limited range and resolution. Stereo from the passthrough cameras works and is unreliable on textureless surfaces. Learned monocular depth is improving quickly and is least accurate at depth discontinuities — which are precisely the edges where occlusion is decided.

Hands are the critical case. A user's real hand passing in front of a virtual object must occlude it, and users test this constantly and unconsciously. **Hand segmentation is therefore given dedicated treatment in every AR system**, separate from general depth estimation, because getting it right matters more than getting everything else right.

3 Making virtual content belong

- **A virtual object lit differently from the room it sits in looks pasted on** — and far more obviously than the same inconsistency would in a fully virtual scene, because the real lighting is right there for comparison.
- **Estimate ambient light colour and intensity** from the camera feed. This is cheap, requires no scene understanding, and is immediately effective — a virtual object that is warm in a warm room and cool in a cool one already looks substantially more present.
- **Estimate the dominant light direction**, so that shading and shadows fall consistently with the real lighting. Errors here are noticed even by people who cannot say what is wrong.
- **Build an environment map from the passthrough cameras** for reflections on glossy virtual surfaces. A virtual chrome object reflecting the actual room is strikingly convincing.
- **And cast shadows onto the real world. A contact shadow beneath a virtual object does more to make it appear to rest on a real surface than any other single technique** — exactly as Module 02 said of depth cues generally, and it is cheap.

PASSTHROUGH IMAGE QUALITY IS THE CURRENT BINDING CONSTRAINT

Passthrough cameras are low resolution, noisy in dim light, and limited in dynamic range compared with the human eye, which handles a room with a bright window without difficulty.

And the cameras are not located at your eyes. They sit on the front of the headset, several centimetres away, so the captured view must be reprojected to each eye's actual viewpoint — which requires depth, and introduces warping and artefacts at depth discontinuities, particularly around the user's own hands and at the edges of near objects.

The result is a view of the world that is usable and visibly degraded. Reading small text through passthrough is uncomfortable; recognising faces and navigating a room are entirely fine.

This is improving quickly, and it is at present the binding constraint on how long anyone will keep a passthrough headset on — more so than weight, battery, or content.

- **Other people are present.** They can see you gesturing at objects they cannot see, and they have no way to know what you are attending to. This is socially awkward in a way that a closed VR headset — which at least unambiguously signals unavailability — is not.
- **Cameras in social spaces raise privacy questions** that a sealed headset does not. Whether the device is recording is not visible to the people around it, and this has a poor history.
- **Attention is divided** between the real and virtual, with genuine safety consequences — the user is walking around a real environment while partly attending to something else.
- **Content must respect the real environment.** A virtual window placed on a real wall is fine; one floating in a real doorway is not, both aesthetically and because people walk through doorways.
- **And the device is worn in public**, which is a design constraint with a conspicuously poor track record. **The social dimension is a substantial part of why AR adoption has lagged behind the technology**, and treating it as a problem outside engineering is how products fail.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Azuma — A Survey of Augmented Reality (free)	The foundational survey. The registration analysis of §2 is clearer here than in most modern treatments.
Meta — Passthrough and Mixed Reality documentation (free)	The current practice: passthrough APIs, scene understanding, anchors, and depth.
Apple — visionOS human interface guidelines (free)	The most carefully considered published thinking on §3's social and environmental design questions, whatever you make of the device.
Google — ARCore lighting estimation documentation (free)	Practical lighting estimation, including environment map generation from a camera feed.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Use a passthrough headset and assess the image quality honestly: read text at several sizes, judge distances, and note where reprojection artefacts appear.
- 2 Place a virtual object on a real table and walk around it. Record every moment where registration error became visible.
- 3 Measure registration error: place a virtual marker on a real one and photograph the offset from several angles.
- 4 Implement occlusion using the headset's depth API and test it with your own hand. Note where the edges fail.
- 5 Place a virtual object in a brightly lit and a dimly lit room without lighting estimation, then with it, and compare.
- 6 Add a contact shadow under a virtual object resting on a real surface. Judge the improvement.
- 7 Build an environment map from passthrough and apply it to a reflective virtual object.
- 8 Implement a persistent anchor, close the application, and confirm the content returns to the same real-world position.
- 9 Wear a passthrough headset in a room with another person for ten minutes. Write a page on the social experience from both sides.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Compare optical and video see-through on six axes.
- 2 Why can optical see-through not render black, and what follows?
- 3 Why is registration error obvious in AR and invisible in VR?
- 4 What tolerances does registration require, and how does latency contribute?
- 5 Give five requirements of good registration.
- 6 Why is occlusion hard in AR, and why are hands the critical case?
- 7 Name five elements of lighting estimation and say which is cheapest and most effective.
- 8 Why is passthrough quality the current binding constraint?
- 9 Give five social or environmental problems AR has that VR does not.